

Causality

Two Causal Frameworks (Part II)

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Roadmap

- Potential Outcome and its traditional assumptions (SUTVA and Unconfoundedness)
- Structural Causal Model and Causal Effect Identification under SCM (Back-door and Adjustment Criterion)
- Efficient Estimation for Adjustment

Overview

Potential Outcome Framework

Structural Causal Model Framework

More Estimators for Adjustment

Stable Unit Treatment Value Assumption (SUTVA)

- SUTVA consists of two assumptions:
 1. no interference,
 2. no different versions of a treatment
(e.g., different surgery $\mathcal{L}$ by different surgeon $\mathcal{S}$)
- SUTVA connects the observed treatment variable (X) with the causal intervention of interest (x) (also called *consistency*), i.e.,
 - ▶ $X_i = 1 \implies Y_i = Y_i(1)$
 - ▶ $X_i = 0 \implies Y_i = Y_i(0)$
- Equivalently: $Y_i = X_i Y_i(1) + (1 - X_i) Y_i(0)$

Violation of SUTVA: Interference

- The potential outcome $Y_i(x)$ for individual i  may depend on the treatment assignments of **other individuals** .
- **Examples:** Effects of vaccination  on community transmission, peer influence in  or education, contagion in infectious diseases, interactions in social networks, and plot-level treatments in agricultural field trials.
- **Implication:** In the presence of interference, standard assumptions for causal inference may no longer hold, and additional assumptions or frameworks are required (e.g., Rosenbaum 2007; Hudgens and Halloran 2008).
- In such settings, estimating *spillover effects* often becomes a key goal.

Typical Causal Estimands in PO

Estimands are often defined as **differences** or ratios, depending on the application.

- Average treatment effect (ATE):

$$\tau = \mathbb{E}[Y(1) - Y(0)]$$

- Conditional average treatment effect (CATE): conditional on covariates

$$\tau(z) = \mathbb{E}[Y(1) - Y(0) \mid \mathbf{Z} = \mathbf{z}]$$

- Average treatment effect for the treated (ATT):

$$\tau = \mathbb{E}[Y(1) - Y(0) \mid X = 1]$$

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Randomization and Unconfoundedness

- In **randomized experiments**, assignment mechanism is known and controlled by investigators
- Randomization *balances*:
 - ✓ both observed and unobserved covariates:

$$X \perp\!\!\!\perp \mathbf{Z} \quad \text{and} \quad X \perp\!\!\!\perp \mathbf{U}$$

- ✓ potential outcomes, i.e. guarantee **unconfoundedness**

$$X \perp\!\!\!\perp (Y(1), Y(0))$$

Randomization and Unconfoundedness

Under randomization

- ✓ causal effects are identified:

$$P(Y(x)) = P(Y(x) | X = x) = P(Y | X = x), \quad x \in \{0, 1\}$$

- ✓ association implies causation:

$$\begin{aligned}\tau^{\text{ATE}} &\equiv \mathbb{E}[Y(1) - Y(0)] \\ &= \mathbb{E}[Y(1)] - \mathbb{E}[Y(0)] \\ &= \mathbb{E}[Y(1) | X = 1] - \mathbb{E}[Y(0) | X = 0] \\ &= \mathbb{E}[Y | X = 1] - \mathbb{E}[Y | X = 0]\end{aligned}$$

Observational Studies

Most common observational assumptions.

1. No unmeasured confounding:

$$\{Y(1), Y(0)\} \perp\!\!\!\perp X \mid \mathbf{Z}$$

- Conditional on **some covariates**, the treatment X is assumed to be (effectively) randomly assigned.

2. Positivity or overlap:

$$0 < P(X = 1 \mid \mathbf{Z}) < 1$$

- Both treatment and control are possible for every value of $\mathbf{Z}$.

We'll assume $\mathbf{Z}$ is given for now, and revisit how to choose it later on.

Identification of the ATE

- Positivity and no unmeasured confounders will identify the ATE:

$$\begin{aligned}\tau &= \mathbb{E}[Y(1) - Y(0)] \\ &= \mathbb{E}_{\mathbf{Z}} \{ \mathbb{E}[Y(1) - Y(0) \mid \mathbf{Z}] \} \\ &= \mathbb{E}_{\mathbf{Z}} \{ \mathbb{E}[Y(1) \mid \mathbf{Z}] - \mathbb{E}[Y(0) \mid \mathbf{Z}] \} \\ &= \mathbb{E}_{\mathbf{Z}} \{ \mathbb{E}[Y(1) \mid X = 1, \mathbf{Z}] - \mathbb{E}[Y(0) \mid X = 0, \mathbf{Z}] \} \\ &= \mathbb{E}_{\mathbf{Z}} \{ \mathbb{E}[Y \mid X = 1, \mathbf{Z}] - \mathbb{E}[Y \mid X = 0, \mathbf{Z}] \}\end{aligned}$$

Identification of the ATE

- Useful to write the treated and control conditional expectation functions (how the mean of the potential outcomes vary with the covariates):

$$\mu_1(\mathbf{z}) = \mathbb{E}[Y(1) | \mathbf{Z} = \mathbf{z}], \quad \mu_0(\mathbf{z}) = \mathbb{E}[Y(0) | \mathbf{Z} = \mathbf{z}]$$

- Key part of the previous proof:

$$\underbrace{\mu_1(\mathbf{z})}_{\text{counterfactual}} = \mathbb{E}[Y | \underbrace{X=1, \mathbf{Z}=\mathbf{z}}_{\text{observational}}], \quad \mu_0(\mathbf{z}) = \mathbb{E}[Y | X=0, \mathbf{Z}=\mathbf{z}]$$

Note that the individual treatment effect is not identifiable but conditional ATE is.

Regression Estimation of the ATE

- Regression estimators $\hat{\mu}_1(\mathbf{z})$ and $\hat{\mu}_0(\mathbf{z})$
 - ▶ Might be linear or nonlinear models (random forest, XGBoost, NN)
 - ▶ practice: estimate separate regressions in each treatment group.
- Regression estimator of the ATE:

$$\hat{\tau}_{\text{reg}} = \frac{1}{N} \sum_{i=1}^N \hat{\mu}_1(\mathbf{z}_i) - \hat{\mu}_0(\mathbf{z}_i)$$

- Note that Conditional ATE is:

$$\tau(\mathbf{z}) = \mu_1(\mathbf{z}) - \mu_0(\mathbf{z})$$

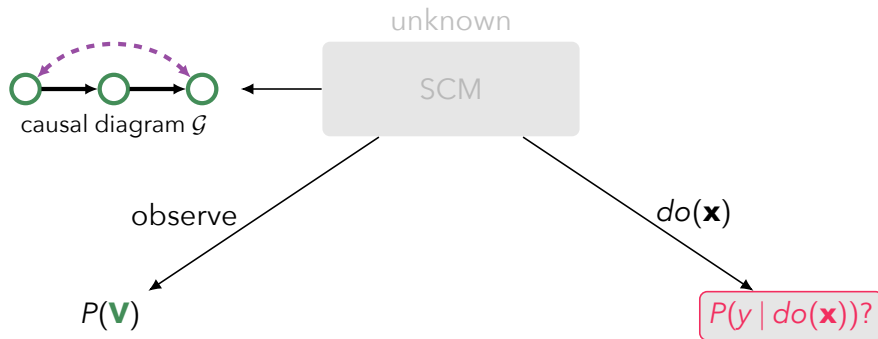
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Causal Inference with SCM



* illustrative purposes

Reading Information from **Causal Diagram** (1/2)

Q: How can we make use of the causal graph $\mathcal{G}$ (the 'view' of model $\mathcal{M}$) in deriving the **causal effect** $P(y \mid do(x))$ from $P(\mathbf{V})$?

In a simple case called Markovian (no bidirected edge):

$$P(\mathbf{v}) = \prod_{v_i \in \mathbf{v}} P(v_i \mid \mathbf{pa}_i)$$

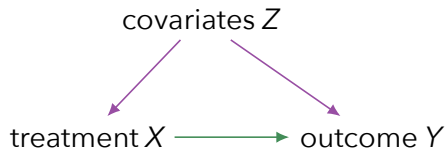
Further, causal effect is given by (still, under Markovian case)

$$P(\mathbf{v} \setminus \mathbf{x} \mid do(\mathbf{x})) = \prod_{v_i \in \mathbf{v} \setminus \mathbf{x}} P(v_i \mid \mathbf{pa}_i)$$

Think about why!? 🤔

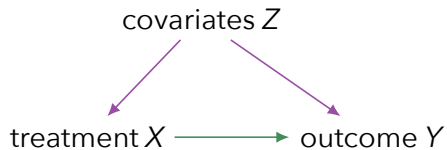
Recall: model is unknown / graph is given.

Truncated Factorization (Markovian Case)



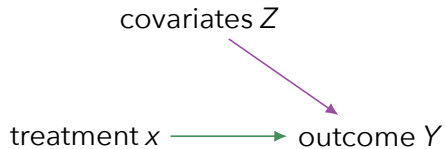
$$P(y, z) = \sum_x P(y, x, z)$$

Truncated Factorization (Markovian Case)



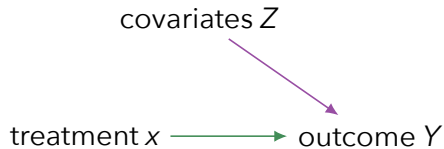
$$P(y, z) = \sum_x P(y \mid x, z) P(x \mid z) P(z)$$

Truncated Factorization (Markovian Case)



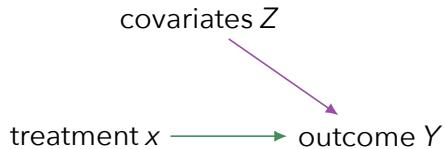
$$P(y, z \mid do(x)) = \sum_{\not x} P(y \mid x, z) \cancel{P(x \mid z)} P(z)$$

Truncated Factorization (Markovian Case)



$$P(y \mid do(x)) = \sum_z \sum_{x \neq z} P(y \mid x, z) \cancel{P(x \mid z)} P(z)$$

Truncated Factorization (Markovian Case)



$$P(y \mid do(x)) = \sum_z P(y \mid x, z)P(z)$$

Reading Information from **Causal Diagram** (2/2)

Q: How can we make use of the causal graph $\mathcal{G}$ (the 'view' of model $\mathcal{M}$) in deriving the **causal effect** $P(y \mid do(x))$ from $P(\mathbf{V})$?

'**Separation**' in graph $\mathcal{G}$ implies '**Conditional Independence**' in P :

$$(X \perp\!\!\!\perp_{\mathcal{G}} Y \mid Z) \implies (X \perp\!\!\!\perp_P Y \mid Z).$$

"**d-separation**" is a method to read **CI** from a given graph.

We can further read d-separation in a manipulated graph (e.g., if intervened).

Back-door Criterion

Association $\approx$ Confounding + Causal



Back-door Criterion

Association $\approx$ Confounding + Causal



Definition (Back-door Criterion)

Find a set $\mathbf{Z}$ such that it can sufficiently explain 'confounding' between $\mathbf{X}$ and $\mathbf{Y}$. Then,

$$P(\mathbf{y} \mid do(\mathbf{x})) = \sum_{\mathbf{z}} P(\mathbf{y} \mid \mathbf{x}, \mathbf{z}) P(\mathbf{z})$$

Back-door Criterion

Association $\approx$ Confounding + Causal



Definition (Back-door Criterion)

A set $\mathbf{Z}$ satisfies the **back-door criterion** wrt a pair $\mathbf{X}, \mathbf{Y}$ in $\mathcal{G}$ **if**;

- (i) no node in $\mathbf{Z}$ is a descendant of $\mathbf{X}$; and
- (ii) $\mathbf{Z}$ blocks ($\sim$ separates) **every path** between $X \in \mathbf{X}$ and $Y \in \mathbf{Y}$ that contains an arrow into X .

Back-door Criterion

Association $\approx$ Confounding + Causal



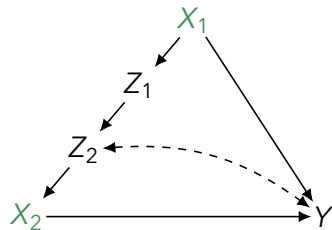
Definition (Back-door Criterion)

A set $\mathbf{Z}$ satisfies the **back-door criterion** with respect to a pair $\mathbf{X}, \mathbf{Y}$ in $\mathcal{G}$ if;

- (i) $\mathbf{Z}$ does *not* disturb **causal paths** from $\mathbf{X}$ to $\mathbf{Y}$; and
- (ii) $\mathbf{Z}$ removes **confounding** between $\mathbf{X}$ and $\mathbf{Y}$.

Adjustment Criterion

- Back-door criterion is not complete for adjustment.
- What is a graphical criterion sound and complete for adjustment?



Definition (Adjustment Criterion)

A set $\mathbf{Z}$ is admissible for adjustment if:

- $\mathbf{Z}$ does not disturb any proper causal path.
 - ▶ No element in $\mathbf{Z}$ is a descendant in $\mathcal{G}_{\overline{\mathbf{X}}}$ of any $W \notin \mathbf{X}$ which lies on a proper causal path from $\mathbf{X}$ to $\mathbf{Y}$.
- All proper non-causal paths in $\mathcal{G}$ from $\mathbf{X}$ to $\mathbf{Y}$ are blocked by $\mathbf{Z}$.

Adjustment Criterion and Unconfoundedness

- **SCM**: Assumptions are encoded in the graph $\mathcal{G}$. Adjustment sets $\mathbf{Z}$ are derived using **graphical criteria**.
- **PO**: Assumptions are stated **declaratively**, e.g.,

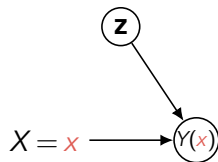
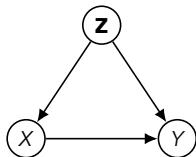
$$Y(x) \perp\!\!\!\perp X \mid \mathbf{Z}.$$

- It is shown that the adjustment criterion (graphical) is equivalent to the unconfoundedness assumption (PO).

“ SCM provides a *transparent* and *flexible* framework: causal assumptions are encoded graphically, enabling systematic identification of adjustment sets and clear reasoning about complex structures. ”

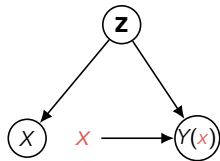
Unconfoundedness as Graphs

Simple Example.



Note: (1) In PO, $Y(x)$ corresponds to Y under $do(x)$ in SCM;
(2) Covariates are unaffected by the intervention, i.e., $\mathbf{Z} = \mathbf{Z}(x)$.

Single-World Intervention Graph



The path $X \leftarrow \mathbf{Z} \rightarrow Y(x)$ is blocked by $\mathbf{Z}$, thus, $Y(x) \perp\!\!\!\perp X \mid \mathbf{Z}$.

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More Estimators for Adjustment

Inverse Probability Weighting (IPW)

Adjustment & Regression-based Estimator

$$P(y \mid \text{do}(X=x)) = \sum_{\mathbf{z}} P(y \mid x, \mathbf{z}) P(\mathbf{z}) = \sum_{\mathbf{z}} \frac{P(y, x, \mathbf{z}) P(\mathbf{z})}{P(x, \mathbf{z})} = \sum_{\mathbf{z}} \frac{P(y, x, \mathbf{z})}{P(x \mid \mathbf{z})}.$$

- Indeed, we can derive

$$\mathbb{E}[Y(1)] = \mathbb{E} \left[\frac{XY}{P(X=1 \mid \mathbf{Z})} \right]$$

- ATE can be identified as (where $e(\mathbf{Z}) = P(X=1 \mid \mathbf{Z})$)

$$\tau^{\text{ATE}} = \mathbb{E} \left[\frac{XY}{e(\mathbf{Z})} - \frac{(1-X)Y}{1-e(\mathbf{Z})} \right]$$

$$\text{FYI, } \mathbb{E} \left[\frac{XY}{e(\mathbf{Z})} \right] = \mathbb{E} \left\{ \frac{1}{e(\mathbf{Z})} \mathbb{E}[XY(1) \mid \mathbf{Z}] \right\} = \mathbb{E} \left\{ \frac{\mathbb{E}[X|\mathbf{Z}]}{e(\mathbf{Z})} \mathbb{E}[Y(1) \mid \mathbf{Z}] \right\} = \mathbb{E} \{ \mathbb{E}[Y(1) \mid \mathbf{Z}] \} = \mathbb{E}[Y(1)]$$

Inverse Probability Weighting (IPW)

Adjustment & Regression-based Estimator

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Doubly Robust Estimator

Q: Can we combine the virtues of **regression** and **IPW** estimator?

Doubly Robust estimator!

consistent if either model is correctly specified; offers protection against model misspecification.

$$\hat{\tau}_{\text{dr}} = \frac{1}{N} \sum_{i=1}^N \left\{ \frac{X_i Y_i}{\hat{e}(\mathbf{Z}_i)} - \underbrace{\frac{X_i - \hat{e}(\mathbf{Z}_i)}{\hat{e}(\mathbf{Z}_i)} \hat{\mu}_1(\mathbf{Z}_i)}_{\text{error correction}} \right\} - \frac{1}{N} \sum_{i=1}^N \left\{ \frac{(1 - X_i) Y_i}{1 - \hat{e}(\mathbf{Z}_i)} + \underbrace{\frac{X_i - \hat{e}(\mathbf{Z}_i)}{1 - \hat{e}(\mathbf{Z}_i)} \hat{\mu}_0(\mathbf{Z}_i)}_{\text{error correction}} \right\}$$

Summary

- Adjustment criterion in SCM and unconfoundedness in PO both aim to justify covariate adjustment for identifying causal effects.
- CATE captures treatment effect conditional on covariates, and ATE is obtained by averaging CATE over the covariate distribution.
- Inverse Probability Weighting (IPW) reweights samples using propensity scores (i.e., $P(X = 1 \mid \mathbf{Z})$) to estimate ATE.
- Doubly robust estimators combine outcome regression and IPW, remaining consistent if either model is correctly specified.